EXPERIMENT 2

AIM: To containerize Application using Docker.

THEORY:

What is Docker?

Docker is an open-source containerization platform used to build, package, deploy, and run applications in isolated environments called containers.

Before Docker, applications were usually installed directly on operating systems. This created problems:

- Different operating systems
- Different library versions
- Dependency conflicts
- "It works on my machine" issues

Docker solves this by packaging an application with all required dependencies into a portable unit called a container.

What is Containerization?

Containerization is a technology that allows multiple applications to run on the same operating system while remaining isolated from each other. A registry stores Docker images. A Docker image is a read-only template used to create containers.

Difference Between Docker Image and Containerization:

Docker image:

A Docker image is a read-only blueprint used to create containers.

It contains everything an application needs:

- Operating system files
- Libraries
- Dependencies
- Application code
- Configuration files
- Environment settings

Containerization:

Containerization is the method of packaging and running applications inside isolated environments called containers. It is a technology concept.

It solves:

- Dependency problems
- Environment differences
- Deployment issues

Docker Image : A packaged template containing the files, libraries, and configuration needed to create a container

Containerization : The technology/process of running applications in isolated containers

The commands implemented are:

1. docker –version : This command checks whether the Docker client (CLI) is installed on your computer. This confirms the Docker CLI installation.

Docker has two main parts:

a) Docker Client

- The command-line tool you type (docker run, docker pull, etc.)
- Sends instructions to the Docker Engine

b) Docker Engine (Daemon)

- Background service that actually creates and manages containers, images, networks, and volumes

2. sc query com.docker.service : sc means Service Control in Windows.This command checks the status of the Docker Desktop Windows service.

3. net start com.docker.service : This starts a Windows service from the command line.Docker Desktop depends on this service to communicate with the Docker Engine.

4. docker version : This checks communication between:

- Docker Client - Shows your installed Docker CLI.
- Docker Server (Docker Engine) - Shows the Docker Engine.

5. docker context ls : Docker contexts define which Docker Engine the Docker CLI communicates with.

6. docker search ubuntu : Searches Docker Hub for available Ubuntu images. Docker Hub is a public image repository.

It contains:

- Official images
- Community images
- Application images

7. docker pull ubuntu : Downloads a Docker image from Docker Hub to your local machine.An image contains:

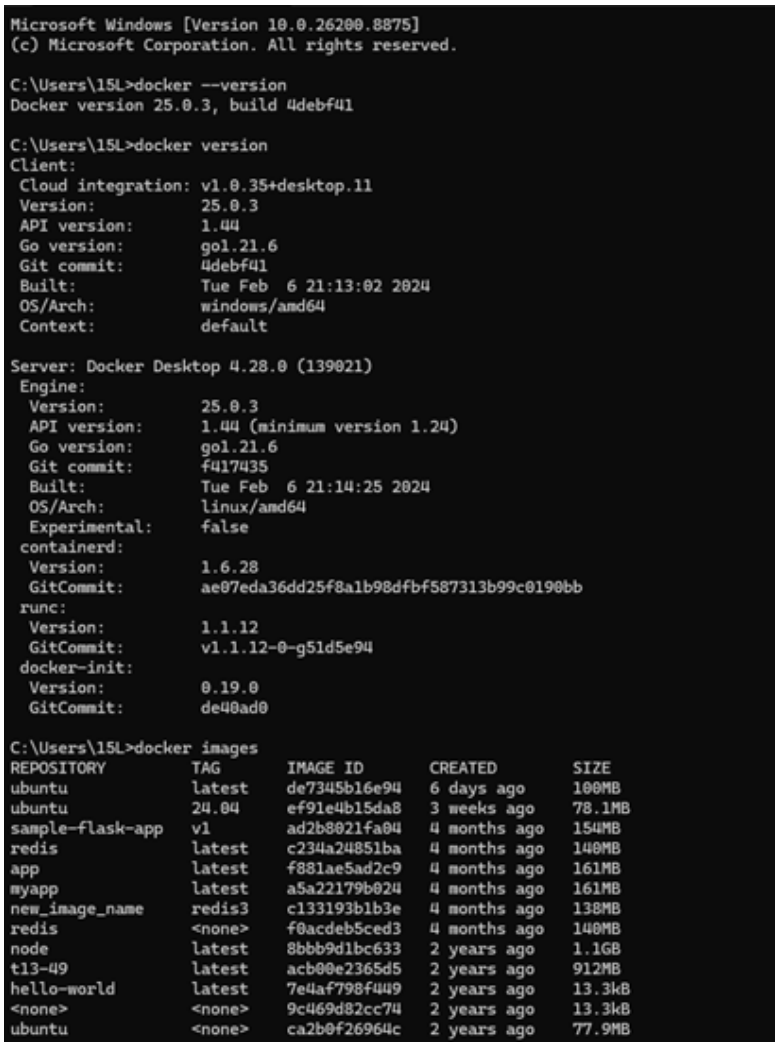
- Base operating system files
- Libraries
- Tools
- Application dependencies

8. docker images : Displays images stored locally. Images are templates used to create containers.

9. docker run -it ubuntu bash : This command creates and starts a container.

10. docker ps : Shows currently running containers.

OUTPUT:



```
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\15L>docker version
Client:
 Cloud integration: v1.0.35+desktop.11
 Version:           25.0.3
 API version:       1.44
 Go version:        go1.21.6
 Git commit:        4debf41
 Built:             Tue Feb  6 21:13:02 2024
 OS/Arch:           windows/amd64
 Context:           default

Server: Docker Desktop 4.28.0 (139021)
 Engine:
  Version:          25.0.3
  API version:      1.44 (minimum version 1.24)
  Go version:       go1.21.6
  Git commit:       f417435
  Built:            Tue Feb  6 21:14:25 2024
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          1.6.28
  GitCommit:        ae07eda36dd25f8a1b98dfbf587313b99c0190bb
 runc:
  Version:          1.1.12
  GitCommit:        v1.1.12-0-g51d5e94
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0

C:\Users\15L>docker images
REPOSITORY         TAG       IMAGE ID       CREATED        SIZE
ubuntu             latest    de7345b16e94   6 days ago     100MB
ubuntu             24.04     ef91e4b15da8   3 weeks ago    78.1MB
sample-flask-app   v1        ad2b8021fa04   4 months ago   154MB
redis              latest    c234a24851ba   4 months ago   140MB
app                latest    f881ae5ad2c9   4 months ago   161MB
myapp              latest    a5a22179b024   4 months ago   161MB
new_image_name     redis3    c133193b1b3e   4 months ago   138MB
redis              <none>    f0acdeb5ced3   4 months ago   140MB
node               latest    8bbb9d1bc633   2 years ago    1.1GB
t13-49             latest    acb00e2365d5   2 years ago    912MB
hello-world        latest    7e4af798f449   2 years ago    13.3kB
<none>             <none>    9c469d82cc74   2 years ago    13.3kB
ubuntu             <none>    ca2b0f26964c   2 years ago    77.9MB
```

```
C:\Users\15L>sc query com.docker.service

SERVICE_NAME: com.docker.service
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE              : 1  STOPPED
        WIN32_EXIT_CODE    : 0  (0x0)
        SERVICE_EXIT_CODE  : 0  (0x0)
        CHECKPOINT         : 0x0
        WAIT_HINT          : 0x0

C:\Users\15L>net start com.docker.service
The Docker Desktop Service service is starting.
The Docker Desktop Service service was started successfully.
```

```
C:\Users\15L>sc query com.docker.service

SERVICE_NAME: com.docker.service
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE              : 4  RUNNING
                             (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE    : 0  (0x0)
        SERVICE_EXIT_CODE  : 0  (0x0)
        CHECKPOINT         : 0x0
        WAIT_HINT          : 0x0

C:\Users\15L>docker context ls
NAME              TYPE              DESCRIPTION                               DOCKER ENDPOINT                                KUBERNETES ENDPOINT   ORCHESTRATOR
default *         moby              Current DOCKER_HOST based configuration   npipe:////./pipe/docker_engine
desktop-linux     moby              Docker Desktop                            npipe:////./pipe/dockerDesktopLinuxEngine

C:\Users\15L>docker context show
default
```

```
redis               <none>    d1397258b209   2 years ago    138MB
redis               <none>    170a1e90f843   2 years ago    138MB
hello-world         <none>    d2c946e258dcb  3 years ago    13.3kB

C:\Users\15L>docker search ubuntu
NAME                      DESCRIPTION                                     STARS     OFFICIAL
ubuntu                    Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid              Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx              Nginx, a high-performance reverse proxy & we…   141
ubuntu/kafka              Apache Kafka, a distributed event streaming …   62
ubuntu/bind9              BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2            Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus         Prometheus is a systems and service monitori…   80
ubuntu/jre                Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/zookeeper          ZooKeeper maintains configuration informatio…   15
ubuntu/dotnet-aspnet      Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/postgres           PostgreSQL is an open source object-relation…   45
ubuntu/python             A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql              MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-runtime     Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis              Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps        Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana            Grafana, a feature rich metrics dashboard & …   12
ubuntu/jdk                Ubuntu rock for OpenJDK                         0
ubuntu/memcached          Memcached, in-memory keyvalue store for smal…   7
ubuntu/rabbitmq           Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir              Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki               Grafana Loki, a log aggregation system like …   2
ubuntu/alertmanager       Ubuntu Rock for Alertmanager. It handles ale…   1
ubuntu/traefik            Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0
```

```
C:\Users\15L>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

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ubuntu/grafana-agent      Ubuntu ROCK for Grafana Agent, an open-sourc…   0

C:\Users\15L>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a1359db865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\15L>docker pull ubuntu:24.04
24.04: Pulling from library/ubuntu
ca2678b20700: Pull complete
Digest: sha256:4fbb8e6a8395de5a7550b33509421a2bafbc0aab6c06ba2cef9ebffbc7092d90
Status: Downloaded newer image for ubuntu:24.04
```

```
ca2678b20700: Pull complete
Digest: sha256:4fbb8e6a8395de5a7550b33509421a2bafbc0aab6c06ba2cef9ebffbc7092d90
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04

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C:\Users\15L>docker images
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ubuntu              24.04     ef91e4b15da8   3 weeks ago     78.1MB
sample-flask-app    v1        ad2b8021fa84   4 months ago    154MB
redis               latest    c234a2485lba   4 months ago    140MB
app                 latest    f881ae5ad2c9   4 months ago    161MB
myapp               latest    a5a22179b024   4 months ago    161MB
new_image_name      redis3    c133193b1b3e   4 months ago    138MB
redis               <none>    f0acdeb5ced3   4 months ago    140MB
node                latest    8bbb9d1bc633   2 years ago     1.1GB
t13-49              latest    acb00e2365d5   2 years ago     912MB
hello-world         latest    7e6af798f049   2 years ago     13.3kB
<none>              <none>    9c469d82cc74   2 years ago     13.3kB
ubuntu              <none>    ca2b0f26964c   2 years ago     77.9MB
redis               <none>    d1397258b209   2 years ago     138MB
redis               <none>    170a1e90f843   2 years ago     138MB
hello-world         <none>    d2c94e258dcb   3 years ago     13.3kB

C:\Users\15L>docker run -it ubuntu bash
docker ps
docker ps -a
docker stop <container_id>
docker rm <container_id>
docker rmi ubunturoot@84f7f3405707:/# docker ps
bash: docker: command not found
root@84f7f3405707:/# docker ps -a
bash: docker: command not found
root@84f7f3405707:/# docker stop <container_id>
bash: syntax error near unexpected token 'newline'
root@84f7f3405707:/# docker rm <container_id>
bash: syntax error near unexpected token 'newline'
root@84f7f3405707:/# docker rmi ubuntu
bash: docker: command not found
root@84f7f3405707:/#
root@84f7f3405707:/# docker pull ubuntu:24.04
bash: docker: command not found
root@84f7f3405707:/# docker run -it ubuntu bash
bash: docker: command not found
root@84f7f3405707:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
```

CONCLUSION:

This experiment helped in understanding the fundamentals of Docker and containerization by exploring how applications can be packaged and executed in isolated containers. Through the implementation of various Docker commands, we learned how to manage Docker services, download images, create and run containers, and verify their status. Overall, the experiment demonstrated how Docker simplifies application deployment, ensures consistency across different environments, and reduces dependency-related issues.